

Article

Social Commerce in Saudi Arabia: Opportunities and Challenges in a Digital Society

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Abstract: Internet and communication technologies have been growing at a rapid pace, enabling data sharing and many other applications. Online trading and e-commerce are important applications that provide ease of business as well as flexibility for buyers. In this paper, we provide an overview of social commerce (s-commerce) in the context of Saudi Arabia. This paper explains s-commerce, the background of Saudi Arabia, and its digital economy vision. This paper also highlights the importance of social media and networking, as well as novel web technologies for s-commerce. We also highlight the research gap in the analysis of s-commerce potential in the Saudi market. This paper conducts a detailed analysis of s-commerce in Saudi Arabia based on a sequential explanatory research design with a quantitative and qualitative phase. The quantitative phase involves a survey based on 504 responses. The qualitative phase involves interview questions of 25 participants. Based on this study, seven themes (each with two or more factors) to enhance s-commerce in Saudi Arabia were identified. Finally, future opportunities and challenges are also discussed in detail.

Keywords: ecommerce; online; digital; social networking



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1. Introduction

The development of the Internet has brought about new communication technologies that enable people to share data and messages in real-time, which is very useful for applications such as E-commerce, or trading over the Internet. This involves financial transactions conducted through the web or mobile apps to buy and sell products or services [1].

With the advancement of technology, businesses have been able to create new sales channels through e-commerce websites [2]. Both businesses and customers have benefited from this. For customers, e-commerce offers the convenience of buying products anytime and from anywhere [2], access to detailed product information [3], flexibility and availability of products [4], and easy price comparisons from different websites [5,6].

Businesses also gain various advantages from e-commerce, including new international sales channels [5], immediate cost savings, improved customer service, a competitive edge by reducing the time from production to sale [7], and increased brand awareness [8].

In recent years, social networking (SN) and content over social media have become widely popular [9]. As of April 2022, there were 4.65 billion SN users worldwide, with 26.6% using these platforms to find products to buy [10]. Companies can use this access to customers' information to tailor their business and marketing strategies based on consumer interests and preferences [11]. SN not only allows users to communicate and share information but also provides new opportunities for buying and selling products online [12].

The growing popularity of SN e-commerce forms the basis of social commerce (s-commerce). S-commerce is one subset of e-commerce, that utilizes platforms of SN as

a gateway to e-commerce, thus facilitating online trade [12–14]. This concept has been evolving since its inception in 2005 [15].

The rise of social commerce has changed the way customers participate in online transactions. They are now more involved, engaging in marketing, comparing, buying, and selling in online stores [16]. Social commerce plays a crucial part in economic growth by providing new ways for businesses and customers to interact [17]. Companies have started using social networks to enhance data sharing and teamwork to boost strategies related to business, including integrating social networks into their branding and marketing activities [18].

Saudi Arabia is experiencing new and exciting developments led by Mohammed bin Salman (Crown Prince) and King Salman. The related changes include new budgets, regulations, and policies aimed at reducing the country's dependence on oil and developing an economy that is more sustainable and depends on diverse factors [19]. This new development started with the launch of "Vision 2030", which aims to increase the contribution of other industries (besides the oil sector) to the GDP by 25% [19]. In this vision, medium-sized enterprises (SMEs) have also been identified as an important part of the economy through jobs and innovation. Additionally, it focuses on localizing specialized industries like the digital economy [20,21].

Saudi Arabia values the use of social e-commerce and promotes online trading mechanisms and platforms [22]. The Saudi Arabian Communication and Information Technology Commission (CITC) has also reported that such policies will be a key step toward the Vision 2030 of a strong economy [23].

Vision 2030 also aims to improve job opportunities for the female workforce of the country [24]. The plan emphasizes developing resources and programs to help the female workforce and young population to develop better employment-related skills. The unemployment rate in Saudi Arabia is expected to decrease by 3.6%, and female employment opportunities are projected to increase by 8% [24]. E-commerce, including social commerce, will improve the Saudi economy by providing job opportunities for unemployed individuals. These policies will also promote entrepreneurship and expansion of businesses [25].

This study aims to introduce social commerce and develop a comprehensive social commerce model for Saudi Arabia. Despite a broad range of studies exploring factors influencing s-commerce adoption, many models still lack a comprehensive approach. Specifically, key elements like sustainability have been insufficiently addressed, and no single study has yet integrated all the relevant factors into one model. This creates a gap in the literature that needs to be filled. Our research seeks to bridge this gap by developing an all-inclusive s-commerce adoption model tailored to Saudi Arabia, ensuring alignment with the Vision 2030 objectives for sustainable growth and digital transformation.

One of the main challenges is identifying the most relevant factors influencing customers' attitudes toward adopting social commerce in the country. The research focuses on examining organizational features that are related to social commerce in Saudi Arabia including social, cultural, psychological, content, and design. The research also considers factors related to the sustainability of the use of social commerce such as knowledge, motivation, quality of online platforms, and personal preferences.

To achieve the study's objectives, the following question is addressed to provide critical insights for developing a holistic social commerce model and successfully adopting social commerce in Saudi Arabia:

Research Question (RQ): What factors should be considered when developing a comprehensive social commerce model for Saudi Arabia?

This paper is structured in a clear and logical sequence. Section 2 provides a review of relevant literature, including an overview of Saudi Arabia's profile and the Saudi Vision 2030 initiative. It also offers a background in e-commerce and Web 2.0. The correlation between Web 2.0 and social networking (SN) and its advantages are highlighted. Additionally, it includes a thorough analysis of existing social commerce models. Section 3 identifies the specific gap in the research that this study aims to fill. In Section 4, the methods used in

conducting the study are detailed. Section 5 explains the research findings. Section 6 discusses the significance of the findings and suggests possible directions for future research. The paper concludes with a summary of the key points discussed.

2. Literature Review and Theoretical Background

This section aims to thoroughly review the current literature to understand the complexities involved in the adoption of social commerce and identify the factors that may influence this adoption. It also includes analyses of existing social commerce models and provides a summary of various models.

2.1. Saudi Arabia

In the Middle East, Saudi Arabia (SA) is a large country, with an area exceeding 2 million km². It also has a huge population, exceeding 25 million [26]. Saudi Arabia is a member of both the G20 and the Gulf Cooperation Council (GCC) [27]. In 2021, its economy ranked among the top 20 worldwide and was the largest in the Arab region, reflecting its status as a rapidly advancing nation [28].

Retailers consider Saudi Arabia a prime location for business due to its youthful population and the high number of active social media users. In 2020, individuals aged 15 to 34 made up 36.7% of the total population [29]. Many Saudi companies leverage social media for commercial activities because of its popularity among both consumers and businesses [30].

The number of social media users in Saudi Arabia is rapidly increasing, driven by the use of mobile technology. In 2021, the country had more than 25 million users on social media platforms such as Facebook. Saudi Arabia was ranked fifth among the top 20 countries based on Snapchat user base size in 2022 [31]. Furthermore, the country is attractive to investors, ranking 27th in the global e-commerce market in 2021, with revenues reaching USD 8 billion [32].

To support the growth of e-commerce, the Saudi Ministry of Commerce and Investment introduced the Maroof service, which helps businesses verify their online stores. This initiative benefits all e-commerce participants, fostering trust between sellers and buyers [33]. It also aligns with the broader Vision 2030 objective of enhancing the quality of life for individuals [24].

Saudi Arabia has a unique social commerce landscape compared with other Gulf countries, and government initiatives such as Vision 2030 are playing a vital role in moving towards a digital economy. So, government-led initiatives are the major difference between Saudi Arabia and other Gulf countries. Another important difference is the large youth population of Saudi Arabia, which plays a major role in boosting social commerce.

2.2. The 2030 Vision of Saudi Arabia

Saudi Arabia is making good progress with the help of reforms under the leadership of Crown Prince Mohammed bin Salman and King Salman. These reforms include new budgets, regulations, and policies aimed at reducing the nation's dependence on oil and fostering a more diversified and sustainable economy [19]. The transformation was initiated with the launch of "Vision 2030", a strategic plan introduced by the Crown Prince. A key objective of this vision is to boost the contribution of the non-oil private sector to GDP from 40% to 65% [19]. The plan also highlights the crucial role of small and medium-sized enterprises (SMEs) in driving economic growth through job creation and innovation. Additionally, it prioritizes the localization of specialized industries, such as the digital economy [20]. Ref. [21] mentions the economy generated by digital means as based on three components: relevant infrastructures, digital processes related to business, and transactions related to digital commerce. Saudi Arabia places high importance on e-commerce and social commerce and is making significant strides to promote these activities for economic gains [22]. The Saudi Arabian Communication and Information Technology Commission (CITC) notes that "the development of e-commerce will support the Vision 2030 goals of

creating a diversified economy and establishing an attractive environment for investment, entrepreneurship, and employment” [23].

Another key goal of Vision 2030 is to increase employment opportunities for the female workforce in Saudi Arabia ([24]). The plan emphasizes developing resources and programs for youth and women to enhance their job prospects and skills. The unemployment rate in Saudi Arabia is projected to decline from 11.6% to 7%, while the employment rate for women is expected to rise from 22% to 30% [24]. E-commerce, including social commerce, is anticipated to stimulate Saudi local markets by creating job opportunities for the unemployed. The Saudi government believes that this will foster entrepreneurship and broaden business prospects [25].

2.3. Saudi Arabia's E-Commerce

E-commerce has secured a substantial market share in the Middle East, with Saudi Arabia emerging as the frontrunner. Studies have highlighted the expansion and economic impact of e-commerce in Saudi Arabia, reflecting a growing embrace of this business model [34,35]. Data and research indicate that Saudi Arabia has witnessed significant e-commerce growth, outpacing other Middle Eastern nations [36,37]. According to [38], the widespread use of Internet technology has been a key driver behind this growth. Much of the existing literature on Saudi e-commerce explores the views of both businesses and consumers.

The Saudi government has been instrumental in advancing e-commerce. Several government efforts have been identified to promote e-commerce, including the introduction of government commerce, expanding Internet education opportunities, and increasing Internet accessibility. The authors of [35] concur that the transition to e-government and the rising adoption of e-commerce by SMEs have strengthened Saudi Arabia's position in the Middle East. Ref. [34] also recognizes the government's contribution to the growth of the information technology sector, which has cultivated a supportive environment for e-commerce.

Nevertheless, some researchers, such as [37], argue Saudi e-commerce industry has yet to achieve its full potential. They observe that the government's efforts have largely focused on the digitization of government processes and education, while areas like e-retailing have received less attention. The authors of [34] also note the role of the Commerce Ministry has not been very active in promoting e-commerce in the country. This Committee was supposed to facilitate the development of e-commerce through education and training, security measures, payment gateways, and IT infrastructure. However, there is limited information on the Committee's activities and the role of the Saudi Ministry of e-commerce. Despite its leadership in the region, Saudi Arabia has yet to fully capitalize on its e-commerce potential.

2.4. Statistics Related to E-Commerce in Saudi Arabia

Saudi Arabia is currently ranked as the 25th largest e-commerce market globally [32]. Numerous statistical studies have shown a growing trend of consumers using the Internet to research and buy things online. In 2021, more than 2 billion users made online purchases. Additionally, sales of global e-retail will reach more than USD 5 trillion by 2022.

In Saudi Arabia, e-commerce revenue for the 2020 fiscal year was reported at USD 7 billion, reflecting a 34% growth compared with the previous year [32]. A report highlights significant trends in Saudi Arabia's e-commerce sector, noting increases of 279% in value and 347% in national debit card transactions. Empirical studies, including those by [35,36], further, confirm the steady rise of e-commerce in the country. Specifically, ref. [35] pointed out that the COVID-19 pandemic significantly boosted e-commerce transactions, alongside factors like government support, perceived ease of use, and convenience, which also contributed to the increased adoption of e-commerce.

However, despite this rapid growth, [36] reported that only 9.0% of businesses in Saudi Arabia have adopted e-commerce, indicating a need for broader and faster adoption to fully

realize the country's potential. This view is supported by [37], who observed that although Saudi Arabia has some of the highest online sales in the Middle East, the country has yet to reach its full potential. Challenges like comfort with real cash payments, concerns over having no control over the quality of the product, and cultural issues have hindered further growth. Nonetheless, the adoption of e-commerce in Saudi Arabia could be accelerated through increased services of ICT, continued support of the government, and ease of use of new technologies.

2.5. Social Networking (SN)

Social networking (SN) involves using social media-related applications that allow one to connect with colleagues and family. SN enables businesses to enhance their social commerce by fostering positive connections with a broader audience in the market. Additionally, SN provides valuable market research opportunities, increases revenue through advertising, and can help build a strong brand due to its extensive market reach [39]. Nowadays, SN not only serves as a platform for users to communicate and share, but it also creates an avenue for online trade [12].

2.6. Web 2.0

Web 2.0 is the second version of the Internet that focuses on allowing people to create and share content easily. Unlike the older version of the web, where websites were mostly static, Web 2.0 lets users interact with websites and each other. This includes features like blogs, social media platforms, and tools for working together online. It has changed the Internet from just reading information to also participating in it, which has helped create new ways for businesses and online communities to grow.

2.7. Correlation of SN with Web 2.0

The relationship between social networking (SN) and Web 2.0 lies in SN's reliance on Web 2.0 for its functionality [37]. The work in [40] supports [37] view, noting that the technical aspects of Web 2.0 provide the foundation for SN by implementing interactive environments and web standards. Ref. [40] explains that Web 2.0 enables SN to create online communities where users can engage in discussions, collaborate, and share opinions on common interests. This interaction between SN and Web 2.0 facilitates consumer engagement and redefines how businesses attract and establish new customers. It does so through forums, blogs, and social media platforms like Twitter, Facebook, and Instagram, enhancing product reputation and increasing sales margins through viral marketing campaigns [37]. Essentially, Web 2.0 and SN influence user interactions by enabling digital content sharing.

2.8. Saudi Arabia's Current SN Scenario

Social networking (SN) avenues are utilized for exchanging information, learning, teaching, and collaboration. In Saudi Arabia, there has been significant growth in SN and Internet usage, particularly e-commerce. Data from the World Bank shows that the Internet using people in Saudi Arabia has increased over the years. Trends in SN usage reflect similar growth, with around 80% users actively engaging on social media.

While SN benefits the entire population, its impact is particularly notable in areas like entrepreneurship and start-ups [41] and university education [37]. Ref. [41] highlights that Saudi Arabia-based start-ups increasingly use SN for their marketing. Similarly, in education and healthcare, SN is embraced as a tool that enhances interaction, problem-solving, critical thinking, decision-making, and creativity. However, challenges remain, such as negative attitudes, misinformation, ethical concerns, and a lack of clear policies, which hinder the adoption of SN and Web 2.0 technologies in Saudi Arabia [42]. Despite these obstacles, SN usage in Saudi Arabia continues to rise.

2.9. Social Commerce

In recent years, advancements in modern technology, especially Web 2.0, have enabled widespread use of social networking sites (SNSs). The development of Web 2.0 and the rise of SNSs have encouraged people globally to share their business-related interests and activities, both with and without direct personal interactions. SNSs aim to create an online environment where users can connect over shared activities, interests, and experiences. Many individuals now rely on SNSs for online shopping, and these platforms have become vital for small and medium enterprises seeking new customers.

The interactive features of SNSs—such as communication, content sharing, and networking—have led to the emergence of a new form of e-commerce known as social commerce [15]. Social commerce, driven by SNS usage and Web 2.0 technologies, is gaining traction as a means for socializing and sharing commercial information. As SNSs continue to grow in popularity, they enable customers to actively interact with one another, enhancing their product evaluations and aiding in more informed purchasing decisions [15].

2.10. The Evolution of Social Commerce

The focus on the social aspect of e-commerce began in the late 1990s when Amazon and eBay first implemented social features allowing consumers to rate sellers and write product reviews. In 1999, these platforms introduced four key social features: eBay Café, Amazon's Listmania, Feedback Forum, and Purchase Circles, all aimed at enhancing community interaction [43].

The term “social commerce” was coined by Yahoo in 2005, when it merged e-commerce with social functions through its “Yahoo Shoposphere”, introducing a new social shopping feature on its site [15]. As Web 2.0 and social networking (SN) technologies emerged, e-commerce businesses began incorporating these tools to offer a more interactive and social shopping experience.

Social commerce has since become a significant field of study for both industry professionals and academics interested in online technologies and their impact on consumers and businesses [44]. The concept of social commerce was first formally discussed in 2006, and its practical beginning is often marked by the launch of the flowers.com Facebook store in 2009. Figure 1 illustrates the evolution of social commerce.

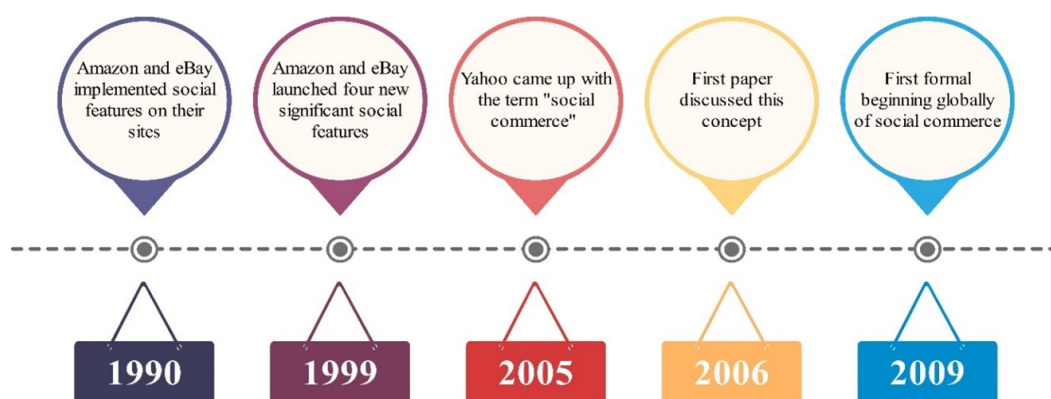


Figure 1. Development of social commerce.

2.11. Benefits of S-Commerce for Business

The advantages of social commerce are closely aligned with those of e-commerce, as social commerce is essentially a subset of electronic commerce that has grown with the rise of social networking sites (SNSs). One key benefit of social commerce is that customers can make informed purchasing decisions based on peer recommendations and find the best prices. Additionally, social commerce enhances customer experiences and boosts a business's competitive edge. It also allows businesses to benefit from spontaneous promotions that occur through SNSs.

Businesses leveraging social commerce continuously add new features and experiences to improve the shopping process. For example, Beyari and Abareshi [45] noted that integrating chatbots, user-friendly product features endorsed by social influencers, broadcasting, and live-selling can significantly boost sales. Chantasombut [46] highlighted that social commerce enhances customer engagement by fostering a two-way interaction between customers and businesses. Moreover, the work in [47] pointed out that social commerce can provide useful statistics related to client behavior which can help improve businesses.

2.12. Review of Current S-Commerce Theories, Frameworks and Models

In this study, holistic social commerce (s-commerce) is developed by integrating factors derived from many models and theories.

2.13. An Integrated Approach and the Initial Factors

Models and theories alone might not offer a complete understanding when studied individually. To address this, we developed a hybrid model by integrating elements from various models and frameworks to gain a more comprehensive perspective. This approach involved a detailed literature review on social commerce to identify the key factors for the initial model. The initial model was then refined based on stakeholder feedback, incorporating themes and factors crucial for successful s-commerce adoption in Saudi Arabia.

We utilized content analysis for the literature review, a widely used method for identifying and analyzing patterns and themes in qualitative data. Qualitative content analysis is described as a method for interpreting text data through systematic classification, coding, and theme identification. There are two main types of content analysis: deductive and inductive. Deductive content analysis tests existing theories using predetermined variables, while inductive content analysis is used to explore data and develop new theories when knowledge is limited.

In this research, we adopted an inductive approach. This technique involves starting with established research aims and questions, rather than a blank slate. We used inductive content analysis to identify the most relevant and significant factors for social commerce and to develop our initial model.

2.14. Research Gap

Several studies have explored social commerce (s-commerce) and the factors influencing customer acceptance, including works by [44,48]. These studies have provided valuable insights into customers' attitudes toward s-commerce. However, despite its practical growth, the theoretical foundation of s-commerce remains fragmented and underdeveloped [15,49]. There is a need for further research to enhance our theoretical and empirical understanding of this evolving field.

In the first stage of this research, a comprehensive literature review was conducted to identify the factors affecting s-commerce adoption. This review included a wide range of sources such as journal articles, conference proceedings, reports, and books. The review revealed that some factors, such as sustainability, were not adequately addressed in existing models. Moreover, no single study has comprehensively covered all relevant factors.

Therefore, this study aims to address these gaps by developing a holistic s-commerce model that integrates all necessary factors for successful implementation in Saudi Arabia, aligning with the country's Vision 2030 goals. The development and evaluation of this model will provide a more thorough understanding of the significant factors influencing consumers' online behavior and the adoption of s-commerce.

2.15. Initial Model

An extensive literature review was conducted to identify factors influencing s-commerce adoption, integrating various insights into an initial model. An integrative review is described as one that "critiques and synthesizes the representative literature on a topic in

an integrative way to generate new frameworks and perspectives on a topic” [50]. This approach was applied to generate a comprehensive model.

An inductive technique for analysis of content was employed to identify and categorize the various s-commerce-related factors. The process involved labeling and grouping these factors into groups, which include the characteristics of the organization, sustainability level, cultural features, and social and psychological behaviors.

These themes collectively form the initial s-commerce model, as illustrated in Figure 2. This model aims to provide a comprehensive understanding of the factors affecting s-commerce adoption, aligning with Saudi Arabia’s Vision 2030 goals.

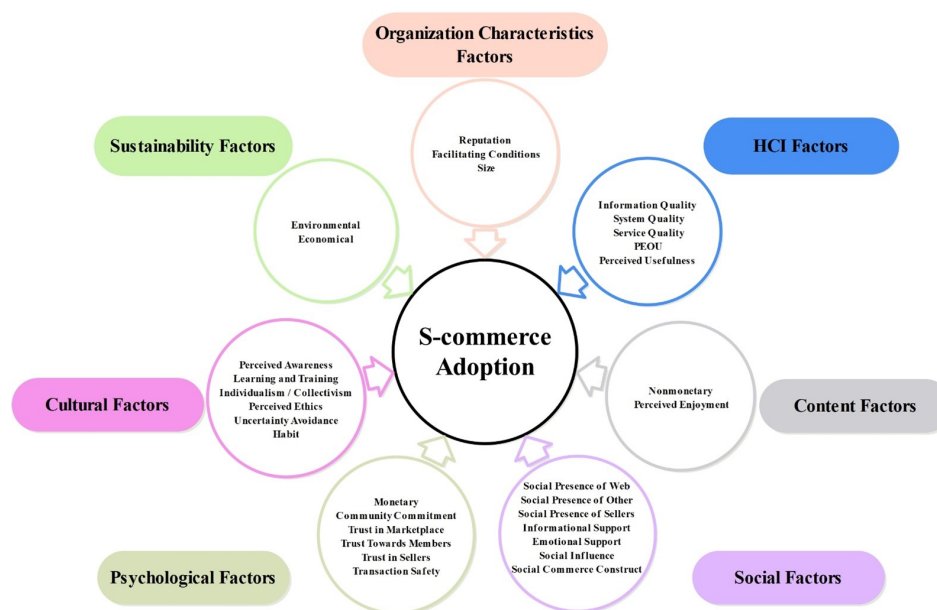


Figure 2. Model for initial social commerce (figure drawn by the author).

3. The Research Methodology and Results

This study used a “sequential explanatory research design” to meet the goals of the research. The used technique involves a couple of stages that focus on the collection of data and analysis of data. The first phase is focused on the quantitative part and the second phase is focused on the qualitative part. According to [51], choosing which method to use first often depends on deciding which method will be the primary one. It is also suggested that if researchers are working with a well-established theoretical foundation but in a new or unclear context, starting with a quantitative study and then following up with a qualitative study can provide deeper insights and address gaps from previous research.

The goal of using the above design is to gain a thorough understanding by analyzing qualitative data to explain and interpret the results from the quantitative study. This design is also useful for triangulation validation, which means using multiple sources and methods to ensure the reliability and accuracy of research findings. In this design, the quantitative phase is often prioritized as the key part of the mixed-methods approach.

3.1. Phase 1: Online Survey (Quantitative)

In this study, an online survey was used first to gather quantitative data. This helped the researcher understand how social network (SN) users respond to social commerce (s-commerce) and identify factors to add to the initial s-commerce adoption model. After collecting this data, the researcher used a qualitative approach to delve deeper into the survey results and better explain the factors identified. This involved conducting semi-structured interviews to verify and refine the s-commerce adoption model, ensuring it was thorough. Using both surveys and interviews fits this study’s pragmatic approach, which supports using mixed methods.

The main goal of the survey conducted online was to confirm the key groups and find which factors should be kept in the model. Exploratory Factor Analysis (EFA) was used to refine the initial s-commerce model. The survey was designed using the Qualtrics XM platform and included Multiple Choice questions (both single and multiple answers), Matrix Table questions (with a five-point Likert scale), and Text Box questions for additional comments.

Given the widespread use and growth of the Internet, online surveys have become a popular and essential method for collecting data [30]. Since this research focuses on a specific aspect of technology adoption, an online survey was an appropriate choice. Participants received a message explaining the survey's purpose, procedures, estimated completion time, and a link to the survey. This message was sent via email and various social networking applications like WhatsApp, Twitter, Facebook, and Instagram.

A total of 1026 responses were collected. The Qualtrics XM platform's Filtering Responses tool was used to keep only those responses from participants who completed the entire survey, resulting in 504 valid responses (49.1%). The data were then downloaded and exported in SPSS (version 26) format. The high number of invalid responses could be due to the survey being too lengthy and covering too many factors, or technical issues such as slow Internet speeds. Feedback indicated that these were significant problems.

Table 1 provides demographic details of participants, including their frequency of social networking (SN) app usage, their s-commerce experiences, and the platform they like. It indicates that the majority of participants were young males (less than 25 years of age). Additionally, a large proportion of participants hold a Post Graduate Diploma. Regarding their employment and income, most work in the government sector and earn over SAR 10,000 (USD 2661) per month.

Table 1. Frequencies and Percentages of Demographic Characteristics and SN Users' Behavior.

Variable	n	%
Gender		
Male	324	64.3
Female	180	35.7
Total	504	100
Age		
18–25	156	30.9
26–30	93	18.5
31–33	80	15.9
34–38	62	12.3
39 and above	113	22.4
Total	504	100
Highest Education Level		
Higher Secondary	79	15.7
Professional Certificate	3	0.6
Diploma	11	2.2
Bachelor's Degree	35	6.9
Post Graduate Diploma	247	49
Master's Degree	98	19.4
PhD	29	5.8
Other	2	0.4
Total	504	100

Table 1. Cont.

Variable	n	%
Occupation Status		
Student	136	27
Unemployed	42	8.3
Employed in the private sector	63	12.5
Employed in the government sector	218	43.2
Self-employed	9	1.8
Retired	29	5.8
Other	7	1.4
Total	504	100
Monthly Income		
Less than SAR 1000 (USD 266)	77	15.3
SAR 1000–3500 (USD 266–931)	101	20
SAR 3501–6000 (USD 932–1597)	21	4.2
SAR 6000–10,000 (USD 1597–2661)	79	15.7
More than SAR 10,000 (USD 2661)	226	44.8
Total	504	100
Usage Duration		
Less than a year	5	1
1–2 years	3	0.6
3–5 years	59	11.7
More than 5 years	437	86.7
Total	504	100
S-commerce Experiences		
Yes	417	82.7
No, but I would consider purchasing in the future	75	14.9
No, and I would not consider purchasing in the future	12	2.4
Total	504	100
Preferred S-commerce Platform (The Participant has the option to select more than one option in this question)		
Facebook	25	4.4
Twitter	57	9.9
Instagram	237	41.4
WhatsApp	134	23.4
Snapchat	120	20.9
Total	573	100

In Table 1, the red numbers highlight the highest percentages for demographic characteristics and SN user behaviors, while the blue numbers show the lowest percentages.

Factor analysis is a statistical technique used to identify how variables in a dataset are related to each other. For this study, seven factors as shown in Figure 2 were identified in

the initial model. Along with these themes, 31 factors were identified from the literature and needed to be evaluated.

Exploratory Factor Analysis (EFA) was employed to explore the underlying factors in the s-commerce model. The first step involved checking if the data were suitable for EFA by evaluating the sample size and the correlation matrix and conducting the tests by Bartlett and Kaiser–Meyer–Olkin (KMO). In the next step, Principal Component Analysis (PCA) was used to pinpoint the factors. To decide how many factors to retain, two methods were applied: examining eigenvalues greater than 1.0 and the screen test. An orthogonal rotation (Varimax) was used to make the results easier to interpret. To ensure strong correlations between variables and factors, a factor loading of 0.6 or higher was required. Additionally, Cronbach’s alpha coefficient (α) needed to be above 0.7 to verify that all variables were consistently measuring the same concept. Table 2 summarizes these procedures and their related measurements.

Table 2. EFA requirements summary.

Step	Values
Determining the suitability of the data	Sample size = 384 Item correlations > 0.30 KMO $\geq$ 0.70 Bartlett’s test (Sig < 0.05)
Determining the factor Extraction method	PCA
Number of the extracted factors	Eigenvalues > 1 Cumulative Percentage of Variance 50% or greater Scree test
Rotational Method	Orthogonal Varimax
Interpretation	Factor loadings > 0.60 Communalities $\geq$ 0.40 Cronbach’s α > 0.7

The first phase of our work considers the results from the online survey, which significantly altered the model’s initial version for Saudi s-commerce, especially regarding a number of factors.

To start, the work assessed whether the data were appropriate for factor analysis by looking at the sample size and adequacy. There were two tests, namely, Bartlett and KMO were performed to evaluate the data’s readiness for factor extraction. The results meet or exceed the minimum standards specified in Table 2, allowing factor analysis to proceed.

The Exploratory Factor Analysis (EFA) and participants’ feedback revealed 7 themes. Each theme had several factors, with each factor comprising various variables. The alpha value of Cronbach for each factor varied between 0.778 to 0.944, surpassing the required standards and showing good internal consistency. For strong connections with the variables, the factors included 4 or more variables having loadings above 0.60.

The first theme, “Organization Characteristics”, includes four factors: “Recognized Company”, “Favorable Reputation”, “Instructions and Guidelines”, and “Delivery”. Within this theme, “Favorable Reputation” had the highest loading at 0.842, while “Offers Well-Known Brand Products” had the lowest loading at 0.618.

The second theme, “Design”, also consists of four factors: “Information Quality”, “System Quality”, “Usability”, and “Perceived Usefulness”. The highest loading for this theme was “Accurate Information” at 0.866, and the lowest was “Understand Specific Requests of Users” at 0.601.

The “Content” theme has two factors: “Non-monetary” and “Perceived Enjoyment”. The variable “Exciting” related to using SN apps for online shopping had the highest loading of 0.888, while “Loyalty Points” had the lowest loading of 0.738.

The fourth theme, “Social”, includes four factors: “Social Presence”, “Sellers’ Informational Support”, “Emotional Support”, and “Stakeholders Endorsement”. The highest

loading within this theme was “Express Interest and Concern in My Welfare” at 0.852, while “Like to See People’s Ratings and Reviews” had the lowest loading at 0.612.

The “Psychological” theme also contains four factors: “Monetary”, “Community Commitment”, “Trust”, and “Transaction Safety”. The highest loading was for “I Feel a Strong Connection to a Company” at 0.884, while “They Keep My Privacy Information Safe” had the lowest loading at 0.624.

The sixth theme, “Culture”, has five factors: “Technical Skills”, “Team Group”, “Perceived Ethics”, “Uncertainty Avoidance”, and “Habit”. The variable “Shopping Using s-commerce Will Cause Me to Lose Privacy of My Payment Information” had the highest loading at 0.870, while “After-Sale Services May Not Be Guaranteed in s-commerce” had the lowest loading at 0.656.

The seventh theme, “Sustainability”, consists of two factors: “Environmental Sustainability” and “Economic Sustainability”. This theme was included due to its absence in previous research and its alignment with Saudi Arabia Vision 2030. The analysis showed that sustainability might impact customers’ attitudes towards s-commerce adoption in Saudi Arabia. For this theme, “Reduces Air Pollution” had the highest loading at 0.862, and “Reduces Paper Bills” had the lowest loading at 0.652.

Table 3 illustrates the initial Saudi s-commerce model and related themes and factors including before and after factor analysis, with all changes highlighted in red.

Table 3. Summary of factor analysis results from the online survey data.

Themes	Before Factor Analysis (Initial S-Commerce Model)	After Factor Analysis
Organization Characteristics	Reputation Size Facilitating Conditions	Favorable Reputation Recognized Company Instructions and Guidelines Delivery
Design	Information Quality System Quality Service Quality PEOU Perceived Usefulness	Information Quality System Quality Usability Perceived Usefulness
Content	Non-monetary Perceived Enjoyment	Non-monetary Perceived Enjoyment
Social	Social Presence of Web Social Presence of Other Social Presence of Sellers Informational Support Emotional Support Social Influence Social Commerce Constructs	Social Presence Sellers’ Informational Support Emotional Support Stakeholders Endorsement
Psychological	Monetary Community Commitment Trust in Marketplace Trust Towards Members Trust in Sellers Transaction Safety	Monetary Community Commitment Trust Transaction Safety
Culture	Perceived Awareness Learning and Training Individualism / Collectivism Perceived Ethics Uncertainty Avoidance Habit	Technical Skills Team Group Perceived Ethics Uncertainty Avoidance Habit
Sustainability	Environmental Economical	Environmental Sustainability Economic Sustainability

3.2. Phase 2: Interview (Qualitative)

Ref. [52] noted that interviews are a widely used method for gathering qualitative data. In this study, interviews aimed to delve into different experiences of s-commerce adoption in Saudi Arabia. The interview data provided a deeper understanding of the quantitative survey results and offered additional explanations for the factors identified, validating improvements to the holistic s-commerce adoption model. The interviews were conducted to ensure that the revised s-commerce model was thorough and to understand why and how these factors are significant for inclusion in the model for Saudi Arabia.

The interview phase followed these four steps, as summarized in Table 4: First, the objectives of the interviews were defined. Next, semi-structured interview questions were developed based on the survey results. In the third step, purposive sampling was used to select experts with s-commerce experience in Saudi Arabia. The fourth step involved conducting the interviews. Finally, the interview data were analyzed using NVivo 12 software.

Table 4. The four steps of the interview phase.

Process	Description
Defining the interview objectives	The interviews are intended for the following purposes: - Refine the enhanced model and determine the factors associated with each theme to design the final holistic s-commerce model for Saudi Arabia. - Confirm the answer to the first research question and respond to the third research question by examining the experts' perceptions.
Formulating the interview questions	semi-structured interview questions were designed based on the online survey results.
Identifying the interviewees	Purposive sampling were used in the third stage to recruit experts who have s-commerce experience in Saudi Arabia.
Conducting the interviews	- Asynchronously semi-structured interviews were conducted via the Qualtrics XM platform. - There were three interview rounds of data collection: nine interviews were conducted in the first round, ten interviews in the second round, and only six interviews were conducted in the last round.

The interviews were conducted with 25 participants from various universities and companies involved in s-commerce in Saudi Arabia. The initial set of interview questions aimed to gather demographic information, including participants' age, gender, education level, job position, years in their current role, and experience with s-commerce technology. The demographic data are presented in Table 5.

Table 5. Participant demographic information.

Demographics Items	Categories	Frequency	Percentage
Gender	Male	15	60%
	Female	10	40%
	Total	25	100%
Age	23–29	0	0%
	30–36	21	84%
	37–43	4	16%
	44–50	0	0%
	50 and above	0	0%
	Total	25	100%

Table 5. Cont.

Demographics Items	Categories	Frequency	Percentage
Highest education level attained	Bachelor degree	6	24%
	Post graduate diploma	0	0%
	Master degree	15	60%
	PhD	4	16%
	Total	25	100%
Years in the current job	Less than one year	0	0%
	1–3 years	2	8%
	4–6 years	13	52%
	7–10 years	8	32%
	More than 10 years	2	8%
	Total	25	100%
Years of experience with social commerce	Less than one year	0	0%
	1–3 years	2	8%
	4–6 years	13	52%
	More than 6 years	10	40%
	Total	25	100%
level of competency in the use of social commerce	Extremely competent	14	56%
	Somewhat competent	8	32%
	Neutral	3	12%
	Somewhat incompetent	0	0%
	Extremely incompetent	0	0%
	Total	25	100%

The analysis of the interview data revealed diverse viewpoints and highlighted the significance of the factors included in the final s-commerce model. Experts confirmed their level of satisfaction with each factor's importance, using a five-point scale ranging from 5 (Extremely Important) to 1 (Unimportant). This scale helped determine the weighting and priority of these factors, as illustrated in Figure 3.

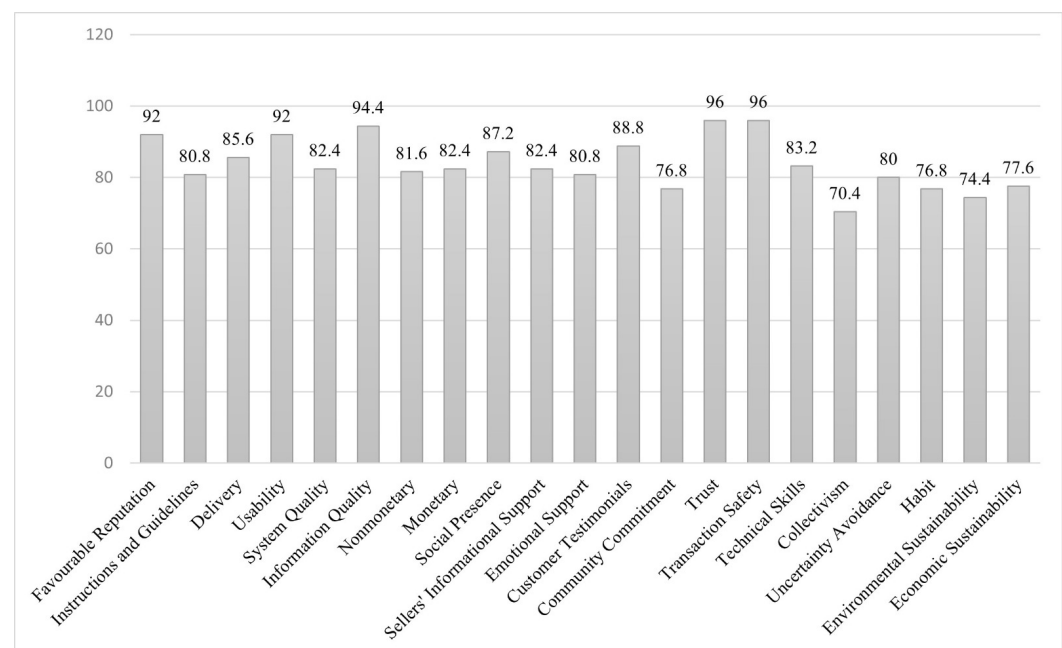


Figure 3. Ratings of overall factors.

The thematic analysis of the interview data revealed seven themes in the improved s-commerce model, each with two or more factors. The first theme, “Organization Charac-

teristics”, includes three factors: “Favorable Reputation”, “Instructions and Guidelines”, and “Delivery”. The “Favorable Reputation” factor received the highest importance score at 92%, while “Instructions and Guidelines” had a lower score of 80.8%.

The second theme, “Design”, consists of four factors: “Usability”, “System Quality”, “Information Quality”, and “Service Quality”. “Information Quality” had the highest importance score of 94.4%, whereas “System Quality” was rated at 82.4%.

The “Perceived Usefulness” theme has two factors: “Non-monetary” and “Monetary”. The “Monetary” factor was slightly more important, with a score of 82.4%, compared with 81.6% for “Non-monetary”.

The fourth theme, “Social”, includes “Social Presence”, “Sellers’ Informational Support”, “Emotional Support”, and “Customer Testimonials”. “Customer Testimonials” was considered the most important with an 88.8% score, while “Emotional Support” received the lowest score of 80.8%.

The “Psychological” theme has three factors: “Community Commitment”, “Trust”, and “Transaction Safety”. Both “Trust” and “Transaction Safety” were rated highest at 96%, while “Community Commitment” had the lowest rating of 76.8%.

The sixth theme, “Culture”, comprises “Technical Skills”, “Collectivism”, “Uncertainty Avoidance”, and “Habit”. “Technical Skills” was rated highest at 83.2%, but “Collectivism” received the lowest rating of 70.4%. Despite its lower score, “Collectivism” was kept in the model due to its important role in s-commerce adoption in Saudi Arabia, a collectivistic society, as noted by Hofstede Insights (n.d.).

Finally, the seventh theme, “Sustainability”, includes two factors: “Environmental Sustainability” and “Economic Sustainability”. “Economic Sustainability” was rated higher at 77.6%, while “Environmental Sustainability” received a score of 74.4%.

Table 6 shows all seven themes and their factors before and after the qualitative analysis, with all changes highlighted in red.

Table 6. Summary of thematic analysis results from the interviews data.

Themes	After Quantitative Data Analysis (Enhanced S-Commerce Model)	After Qualitative Data Analysis (Final Holistic S-Commerce Model)
Organization Characteristics	Favorable Reputation Recognized Company Instructions and Guidelines Delivery	Favorable Reputation Instructions and Guidelines Delivery
Design	Usability System Quality Information Quality Service Quality Perceived Usefulness	Usability System Quality Information Quality Perceived Usefulness
Content	Non-monetary Perceived Enjoyment Monetary	Non-monetary Monetary
Social	Social Presence Sellers’ Informational Support Emotional Support Stakeholders Endorsement	Social Presence Sellers’ Informational Support Emotional Support Customer Testimonials
Psychological	Monetary Community Commitment Trust Transaction Safety	Community Commitment Trust Transaction Safety

Table 6. Cont.

Themes	After Quantitative Data Analysis (Enhanced S-Commerce Model)	After Qualitative Data Analysis (Final Holistic S-Commerce Model)
Culture	Technical Skills Team Group Perceived Ethics Uncertainty Avoidance Habit	Technical Skills Collectivism Uncertainty Avoidance Habit
Sustainability	Environmental Sustainability Economic Sustainability	Environmental Sustainability Economic Sustainability

4. Findings and Discussion

This study identified seven key themes that can enhance the successful adoption of s-commerce in Saudi Arabia: organization characteristics, design, perceived usefulness, social, psychological, cultural, and sustainability. Each theme comprises two or more factors, which are elaborated upon in the subsequent sections.

4.1. Organization Characteristics Theme

This theme covers elements of an organization that can affect customers' perceptions of the company, particularly regarding its capabilities, size, and reputation. The section explores the factors within the organization characteristics theme and their significance for adopting s-commerce and their inclusion in the HSMSA model. The organization characteristics theme and its associated factors were validated through a literature review and data collection, including online surveys and semi-structured interviews with s-commerce experts. As shown in Table 7, this theme consists of three factors: favorable reputation, instructions and guidelines, and delivery.

Table 7. Factors associated with the "Organization Characteristics" theme.

Factor	Brief Summary
Favorable reputation	The degree to which consumers perceive that an organization is reliable and concerned about its clients.
Instructions and guidelines	The extent to which a person's perception that an organizational structure and technical infrastructure are available to facilitate system use.
Delivery	One of the success factors for social commerce firms is having speedy and convenient delivery of products.

4.2. Design Theme

This theme seeks to identify the factors that impact customers' attitudes towards adopting s-commerce, specifically focusing on usability and website quality. The results from the online survey and semi-structured interviews indicate that these aspects are crucial and should be included in the final HSMSA model. As shown in Table 8, this theme includes four factors: usability, system quality, information quality, and service quality.

Table 8. Factors associated with the “Design” theme.

Factor	Brief Summary
Usability	The extent to which persons can use a website to accomplish their goals in an effective, efficient, and satisfactory manner.
System quality	Is associated with the desired and essential capabilities of a website such as its reliability, availability, and response time.
Information quality	Determines the accuracy, completeness, and timeliness of the website content.
Service quality	Indicates consumers’ evaluations of the services and supports provided through the website by the service provider.

4.3. Perceived Usefulness Theme

In the perceived usefulness theme, this section focuses on the factors that influence the adoption of s-commerce, considering both monetary and non-monetary aspects. Given that s-commerce depends heavily on individual behavior, perceived usefulness is a key motivator for individuals to participate in s-commerce activities. The significance of this theme and its factors (monetary and non-monetary) has been supported by the literature review and the data from both quantitative and qualitative analyses. Table 9 provides a brief description of each factor.

Table 9. Factors associated with the “Perceived Usefulness” theme.

Factor	Brief Summary
Monetary	The degree to which a customer concentrates on acquiring low-priced services or products.
Non-monetary	The emotional efforts required, and amount of time spent on a transaction (purchase).

4.4. Social Theme

Social commerce is closely tied to the use of social networks, where a person’s behavior is shaped by social factors that foster collaboration and communication among individuals. This theme primarily examines the experiences, communications, and relationships that affect customers’ attitudes, lifestyles, and personalities. It includes four key factors, as shown in Table 10, social presence, sellers’ informational support, emotional support, and customer testimonials. The data analysis indicated that all four factors are essential and should be incorporated into the final HSMSA model.

Table 10. Factors associated with the “Social” theme.

Factor	Brief Summary
Social presence	Is associated with the ability of a social networking site to elicit a feeling of warmth and sociability.
Sellers’ informational support	Refers to delivering messages that can be useful for problem-solving in the form of advice, knowledge, or recommendation.
Emotional support	Defined as delivering messages that include passionate interests such as caring.
Customer testimonials	“The presence of comments, ratings, and reviews about products” (https://www.overleaf.com/project/66b01e7e600e2c820a1d0604#_ENREF_260 , accessed on 11 July 2024).

4.5. Psychological Theme

When buying products through social network apps, consumers take into account factors like motivation, privacy, and security. These aspects significantly affect a person’s mindset and decision-making process, and they represent the psychological factors examined in this study. The identified psychological factors influencing the adoption of

s-commerce include community commitment, trust, and transaction safety, as outlined in Table 11. Both the online survey and SSI results confirm that these three factors are crucial for s-commerce adoption in Saudi Arabia and should be incorporated into the final HSMSA model.

Table 11. Factors associated with the “Psychological” theme.

Factor	Brief Summary
Community commitment	A psychological bond that depicts a consumer’s connection with a company.
Trust	Is considered to be the key to the successful establishment of a long-term relationship and it becomes even more important in the context of online platforms because much uncertainty surrounds the online environment.
Transaction safety	A social commerce site’s ability to protect its consumers before and after a purchase

4.6. Culture Theme

Globalization has significantly impacted the business world, making culture a critical factor in all business activities. Culture directly influences how consumers adopt new technologies. Cultural factors play a role in the acceptance and adoption of information technologies. This theme, supported by literature and both qualitative and quantitative data, encompasses four factors: technical skills, collectivism, uncertainty avoidance, and habit. Table 12 provides brief definitions of each factor.

Table 12. Frequencies and Percentages of Demographic Characteristics and SN Users’ Behavior.

Factor	Brief Summary
Technical skills	Refer to the practical abilities that customers possess that assist them in using s-commerce.
Collectivism	Refers to the extent to which individuals in a community or an organization desire to be active as an individual or a part of a group.
Uncertainty avoidance	Defined as the degree to which individuals in a community worry about unpredictable cases.
Habit	“The extent to which people tend to perform behaviors (use IS) automatically because of learning” (https://www.overleaf.com/project/66b01e7e600e2c820a1d0604#_ENREF_260 , accessed on 18 October 2024).

4.7. Sustainability Theme

This theme primarily examines how environmental and economic sustainability influence Saudi customers’ willingness to adopt social commerce. As outlined in Table 13, the sustainability theme includes two factors: environmental sustainability and economic sustainability. With the growing green trend, customers are increasingly aware of the need for sustainability to protect the planet. Both the online survey and expert interviews indicated that these aspects of sustainability are significant in Saudi Arabia. The majority of interviewees emphasized the crucial role of environmental and economic sustainability in driving the adoption of social commerce. Some even connected the importance of sustainability to Saudi Arabia’s Vision 2030 initiative. As a result, the sustainability theme was incorporated into the final HSMSA model.

Table 13. Factors associated with the “Sustainability” theme.

Factor	Brief Summary
Environmental sustainability	Described as appropriate engagement with the environment to minimize natural resource depletion or deterioration and to ensure long-term environmental quality.
Economic sustainability	An economy’s capacity to sustain a specific level of economic production for an indefinite period.

The researcher adjusted the enhanced s-commerce model for Saudi Arabia to reflect the new insights, as illustrated in Figure 4.



Figure 4. Holistic S-commerce model for Saudi Arabia (final version).

5. Research Contributions

The key contribution of this research lies in identifying the significant factors influencing customers' attitudes towards adopting s-commerce in Saudi Arabia. This study provides both theoretical and practical contributions to the s-commerce field, particularly within Saudi Arabian sectors.

5.1. Theoretical Contributions

Theoretically, this study adds to the existing literature by highlighting the crucial factors necessary for successfully implementing s-commerce in Saudi Arabia. It addresses the concern over the lack of theoretical models guiding s-commerce implementation in the region. By doing so, the thesis provides new theoretical and academic insights that could benefit stakeholders in research centers, institutions, and universities in Saudi Arabia. This includes scholars, academic staff, and students at the PhD and master's levels, who could leverage s-commerce benefits in their academic activities. This study's comprehensive examination of all potential factors using an exploratory approach enriches the current literature, particularly in developing countries and the Gulf Cooperation Council (GCC) countries.

5.2. Practical Contributions

Practically, the findings offer valuable guidance for companies, organizations, and governments on promoting s-commerce adoption and improving its use in Saudi Arabia. Stakeholders such as companies, managers, and marketers can use the final model to identify critical areas requiring attention for maximizing s-commerce benefits. This study emphasizes the importance of seven themes: organization characteristics, design, perceived usefulness, social, psychological, cultural, and sustainability.

Moreover, this research aligns with the objectives of Saudi Vision 2030, which aims to strengthen the Saudi economy by diversifying income sources and promoting economic growth. This study provides practical recommendations for Saudi Internet entrepreneurs, helping them encourage consumers to adopt s-commerce. This, in turn, supports entrepreneurs in achieving their financial goals and contributes to a stronger economy. Additionally, the holistic model developed in this study could be applicable in other GCC countries, given their shared political, linguistic, cultural, economic, and financial similarities.

6. Conclusions and Future Research

In conclusion, this study has provided a comprehensive analysis of social commerce in Saudi Arabia, highlighting its growing importance in the digital economy. Through a mixed-method approach, we identified seven key themes that can enhance the adoption of s-commerce in the country. These findings contribute to a deeper understanding of the potential and challenges of s-commerce within the Saudi market. Furthermore, this study addresses gaps in existing research and offers valuable insights for future opportunities, aligning with Saudi Arabia's Vision 2030 goals for digital transformation.

The results of this study open up several potential areas for future research on s-commerce. These suggestions are relevant for researchers in both developed and developing countries. For example, this study used a cross-sectional approach to gather data from social network users in Saudi Arabia, but future research should consider a longitudinal approach. This would help us understand how various factors interact over time and affect the adoption of s-commerce.

Additionally, due to challenges like the COVID-19 pandemic and the time zone difference between Australia and Saudi Arabia, this study used asynchronous semi-structured interviews. Future studies could use face-to-face interviews to better interpret body language and facial expressions and to obtain more detailed responses from participants.

Since this study was conducted in Saudi Arabia, the findings might not apply to other regions with different characteristics. Future research could apply the final model to other populations to validate and compare the results.

This study used convenience sampling, which may limit how well the results can be generalized. Convenience sampling can cause a potential bias towards younger and tech-savvy respondents. Future research should use probability sampling methods, such as simple random sampling, to enhance the generalizability and validity of the findings.

Lastly, while this study focused on the factors influencing customers' attitudes towards s-commerce, future research should also look into what encourages businesses and organizations to adopt s-commerce. This would offer a fuller understanding of what drives the success of s-commerce.

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